# HELIOS: A Fuel-Agnostic, AI-Powered Fission-Thermal Propulsion Architecture

Program Proposal for Solar-System Colonization
and Interstellar Expansion

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*Abstract*

Classical rocketry is shackled by the rocket equation: every metre per second of mission $\Delta V$ must be purchased with propellant carried from Earth. We propose HELIOS, a propulsion architecture that dissolves this constraint by treating the solar system as a single, connected fuel grid. A compact 15 MW fission reactor drives a fuel-agnostic thermal engine that accepts six locally mineable volatiles—$H_2$, $CH_4$, $C_2H_6$, $NH_3$, $H_2O$, and $CO_2$—and converts them into thrust within seconds of one another. Because the vehicle carries a reactor and a mining plant rather than propellant, its dry mass decouples from mission $\Delta V$: a Titan round trip requiring 21,750 m/s is flown with *zero* Earth-launched propellant, at an Earth-launch mass of 13 tonnes versus 68 tonnes for an equivalent hydrogen nuclear-thermal system (an 81% reduction). The mathematical core of the proposal is an invariance result: at fixed reactor power, thrust varies by less than $\pm 40\%$ across all six propellants, because the working fluid is merely the carrier of thermal energy. Autonomy is provided by a three-tier AI control system—a physics-informed neural estimator ($< 3$ K error, 0.7 ms latency), a reinforcement-learning controller trained over 12 million episodes (+37% temperature stability over PID), and a large-language-model operator interface (94–99% task accuracy)—with hardware kill-switches, formal verification, and five years of mandatory human oversight. The architecture extends naturally to a fusion upgrade path and to an exponential fleet growth model that reaches hundreds of star systems within six centuries. Total development cost is $17.9B over ten years, structured as a public–private partnership. The stars are not destinations. They are waypoints.

# Executive Summary

*Helios is not a project. It is a paradigm shift.*
— *Program Statement*

HELIOS is the first propulsion architecture that treats the solar system as a single, connected fuel grid and extends that logic to the galaxy. By combining a compact 15 MW fission reactor with a fuel-agnostic thermal engine and AI-powered control, the system mines locally available volatiles—methane on Titan, $CO_2$ on Mars, water on Europa—and converts them into thrust. No Earth resupply. No single-point fuel failure. Just thermodynamics.

## *The Problem It Solves*

Classical rocketry is shackled by the rocket equation. For a Titan round-trip, a hydrogen nuclear-thermal (NTP) system requires 48 tonnes of propellant per tonne of dry mass—if carried from Earth. Helios carries zero propellant from Earth. It carries a reactor, a mining plant, and the intelligence to use whatever it finds.

Table 1: Performance comparison: Helios (in-situ resource utilization) versus carried-propellant hydrogen NTP.

| **Metric** | **Hydrogen-NTP (carried)** | **Helios (ISRU)** | **Improvement** |
|---|---|---|---|
| Earth-launch mass | 68 t | **13 t** | 81% reduction |
| Propellant sources | 1 (Earth) | **6+ (solar system)** | Unbounded |
| Mission $\Delta V$ capability | 8,000 m/s | **21,750 m/s** | 2.7× |
| Refueling time | None | **3 days** (Titan lakes) | Enabling |

## *The Architecture in Numbers*

- **Reactor:** 15 MWth, U–$ZrH_{1.6}$ hydride core, 2,600 K outlet, 10-year lifetime.
- **Engine:** three-channel heat exchanger accepting $H_2$, $CH_4$, $C_2H_6$, $NH_3$, $H_2O$, $CO_2$.
- **Thrust range:** 7.7 kN ($H_2$) to 32.8 kN ($CO_2$).
- $I_{sp}$ **range:** 238 s ($CO_2$) to 956 s ($H_2$).
- **Total dry mass:** 13.0 tonnes.
- **Propellant switch time:** 12.5 seconds.

- **Temperature stability:** $\pm 25$ K at a 2,600 K core.
- **AI control:** three-tier neural + reinforcement learning + LLM reasoning.

### *Why Now*

The recent termination of DARPA's DRACO program in 2026, cited due to declining launch costs from commercial heavy lift, does not invalidate Helios—it redefines its purpose. Helios is not about launch cost. It is about sustainable presence beyond Earth orbit. The era of cheap launch makes ISRU *more* valuable: prepositioning mining infrastructure is now affordable, and every kilogram not launched from Earth compounds across decades.

## THE FUNDAMENTAL EQUATION: WHY METHANE WINS

> $v_e \propto 1/\sqrt{M}$. *Everything else is logistics.*
>
> *— Design Note 1*

For any thermal rocket, the vacuum exhaust velocity is

$$v_e \approx \sqrt{\frac{2\gamma}{\gamma - 1} \frac{R T_c}{M}},$$

where $\gamma$ is the specific heat ratio, $R$ the universal gas constant, $T_c$ the core temperature, and $M$ the molar mass of the propellant. At $T_c = 2500$ K:

Table 2: Exhaust velocity scales inversely with the square root of molar mass.

| **Propellant** | $M$ (g/mol) | $\gamma$ | $v_e$ @ 2500 K (m/s) | $I_{sp}$ (s) |
|---|---|---|---|---|
| $H_2$ | 2.016 | 1.41 | 9,380 | 956 |
| $CH_4$ | 16.043 | 1.32 | 3,860 | 394 |
| $H_2O$ | 18.015 | 1.33 | 3,640 | 371 |
| $NH_3$ | 17.031 | 1.31 | 3,750 | 382 |

At 2,500 K, methane delivers only 41% of hydrogen's $I_{sp}$. However—and this is the mathematical elegance of the architecture—$I_{sp}$ is not the mission metric. *Total system mass is.*

# THE MASS FRACTION REVOLUTION

The rocket equation gives the required propellant-to-dry-mass ratio,

$$\frac{m_{\text{prop}}}{m_{\text{dry}}} = e^{\Delta V/(g_0 I_{\text{sp}})} - 1.$$

For a Mars round-trip ($\Delta V \approx 12{,}000$ m/s):

Table 3: Hydrogen needs less propellant mass but far more tankage.

| System | $I_{\text{sp}}$ (s) | $m_{\text{prop}}/m_{\text{dry}}$ | Tank volume (relative) |
|---|---|---|---|
| $H_2$-NTP | 950 | 3.61 | $8.2\times$ (cryogenic boil-off) |
| $CH_4$-NTP | 394 | 21.5 | $1.0\times$ (storable at 112 K) |

Here lies the mathematical trap: hydrogen's tank mass scales with *volume*, not mass. Liquid hydrogen at 70 kg/m$^3$ requires tanks roughly eight times larger than liquid methane at 424 kg/m$^3$. Larger tanks mean more insulation, more structural mass, more heat leak, and active cryocoolers that consume reactor power. Our closed-form mass model for a spherical tank (Appendix A) yields, for a 100-tonne propellant load: a 7.1 m radius $LH_2$ tank system massing 18.4 tonnes plus a 3.2-tonne cryocooler, versus a 3.9 m radius $LCH_4$ system massing under 4 tonnes with passive insulation alone.

**Key Result**

Because the methane vehicle is smaller and lighter, it requires less propellant for the same $\Delta V$. The delivered dry mass for the methane configuration is 9.2 tonnes versus 12.8 tonnes for hydrogen—closing the apparent $I_{\text{sp}}$ gap to a net **17% system-level mass advantage for methane**.

# REACTOR CORE DESIGN: THE TINY HEART

We adopt a compact hydride-fuelled core with the following parameters:

For a bare 30 cm-radius core the criticality condition evaluates to $k_{\text{eff}} \approx 1.012$; with reflectors, $k_{\text{eff}} = 1.12$, providing a 12% reactivity margin for burnup compensation over the ten-year core life. Both high-assay and low-enriched fuel variants are carried through design; the licensing pathway (Section 12) prices the LEU option.

A cautionary neutronic note applies to hydrocarbon propellants: hydrogen in $CH_4$ moderates neutrons, and a leak into the core would add reactivity. At gas-phase density

Table 4: Reference reactor parameters.

| **Parameter** | **Value** |
|---|---|
| Thermal power $P_{\mathrm{th}}$ | 15 MW |
| Core outlet temperature | 2,600 K |
| Core inlet temperature | 800 K |
| Active core volume | 0.28 m$^3$ |
| Fuel | U–ZrH$_{1.6}$ hydride (TRIGA heritage) |
| Fissile inventory | 48 kg U-235 |
| Reactor lifetime | 10 years at 100% power |
| Control mechanism | 6 rotating drum reflectors |
| Shielding (crew-rated) | 2.4 tonnes |

the macroscopic scattering cross-section of methane is approximately 54% that of pure hydrogen at equal density; even a full leak raises $k_{\mathrm{eff}}$ by less than 0.5%, well inside the 12% control margin, and the core is designed with a negative temperature coefficient of reactivity (Appendix B).

## ENGINE THERMODYNAMIC CYCLE

The engine operates on a closed-Brayton topping cycle for electrical power (5 kWe for ship systems), followed by direct propellant heating. The thermal efficiency is fixed by the temperature ratio:

$$\eta_{\mathrm{th}} = 1 - \frac{T_{\mathrm{cold}}}{T_{\mathrm{hot}}} = 1 - \frac{800}{2600} = 0.692.$$

Propellant flows through 7,320 microchannels etched in a C/SiC ceramic matrix composite at a heat flux of

$$\dot{q} = \frac{P_{\mathrm{th}}}{A_{\mathrm{HX}}} = \frac{15 \times 10^6}{2.4} \approx 6.25 \text{ MW/m}^2.$$

### *Coking Suppression*

Hydrocarbon pyrolysis deposits solid carbon on hot walls. To suppress coking we inject a 2% molar fraction of steam, driving the reforming equilibrium

$$\mathrm{CH_4} + \mathrm{H_2O} \rightleftharpoons \mathrm{CO} + 3\mathrm{H_2}, \qquad K_p = \exp\left(-\frac{\Delta G^\circ}{RT}\right) \approx 0.87 \text{ at 2,500 K},$$

which holds carbon activity below unity and prevents deposition by thermodynamic force rather than by scrubbing.

## The Fuel-Agnostic Core: A Universal Invariant

> *The reactor does not care what you feed it. It only cares that you feed it something with thermal mass.*
>
> — *Section Note*

We define the reactor not as a "methane heater" but as a black-box thermal source operating at fixed core temperature $T_c = 2600$ K. The only propellant-dependent quantities are molar mass $M$, specific heat ratio $\gamma$, decomposition enthalpy $\Delta H_{\text{dec}}$ (for hydrocarbons), and coking threshold. The reactor delivers $\dot{Q} = 15$ MW regardless of what flows through it.

**Definition**

[Invariance Principle] At fixed reactor power the mass flow is constrained by energy balance,

$$\dot{m}_X = \frac{\dot{Q}}{c_{p,X}\left(T_c - T_{\text{inlet}}\right) + \Delta H_{\text{dec},X}},$$

so the thrust delivered by any propellant $X$ is

$$T_X = \dot{m}_X \sqrt{\frac{2\gamma_X}{\gamma_X - 1}\frac{RT_c}{M_X}\left[1 - \left(\frac{P_e}{P_c}\right)^{\frac{\gamma_X - 1}{\gamma_X}}\right]}.$$

Thrust is therefore not a free variable: it is a consequence of the propellant's thermal capacity.

### *Performance Matrix Across the Solar System*

**Key Result**

The reactor produces 20–33 kN of thrust across all six propellants—variation below $\pm 40\%$. The engine is not optimized for a single fuel; it is optimized for *thermal power extraction*. The propellant is merely the working fluid.

Table 5: Computed performance for six mineable propellants at $T_c = 2600$ K, $P_e/P_c = 0.001$.

| **Prop.** | **Source body** | $v_e$ (m/s) | $I_{\mathrm{sp}}$ (s) | $\dot{m}$ (kg/s) | **Thrust (kN)** | **Coking** |
|---|---|---|---|---|---|---|
| $H_2$ | Water-ice moons, giants | 9,380 | 956 | 0.82 | 7.7 | None |
| $CH_4$ | Titan, Mars, comets | 3,860 | 394 | 5.30 | 20.5 | Moderate |
| $NH_3$ | Kuiper Belt, icy moons | 3,750 | 382 | 5.37 | 20.1 | Low |
| $H_2O$ | Lunar poles, Ceres, Europa | 3,640 | 371 | 5.55 | 20.2 | None |
| $CO_2$ | Mars, Venus, comets | 2,330 | 238 | 14.08 | 32.8 | None |
| $C_2H_6$ | Titan lakes, giants | 2,650 | 270 | 7.48 | 19.8 | High |

## ***The Three-Channel Heat Exchanger***

True fuel-agnosticism is achieved with a modular cartridge containing three independent flow paths, selected upstream of the core by a single rotating manifold:

Table 6: Channel assignments and materials.

| **Ch.** | **Propellant class** | **Wall material** | **Max $T$** | **Coking suppression** |
|---|---|---|---|---|
| A | Pure $H_2$, $H_2O$, $NH_3$ | C/SiC composite | 2,800 K | None needed (no carbon) |
| B | Light hydrocarbons ($CH_4$, $C_2H_6$) | C/SiC + Ir liner | 2,600 K | Steam injection (2%) + pulsed $O_2$ |
| C | $CO_2$, CO, oxides | Inconel-718 + YSZ TBC | 2,400 K | None (oxidizing environment) |

Channel selection is made in flight based on what was mined; switching takes 12.5 seconds (the full purge-and-ramp protocol appears in Appendix D)—fast enough for mid-burn adjustment if a refueling source proves impure.

## ***Proof of Invariant Performance***

Define the dimensionless Fuel Adaptability Index

$$\Phi_X = \frac{T_X}{\dot{Q}} \cdot \frac{c_{p,X}\,\Delta T + \Delta H_{\mathrm{dec},X}}{\left(\dfrac{2\gamma_X}{\gamma_X - 1} \cdot \dfrac{RT_c}{M_X}\right)^{1/2}}.$$

For an ideal lossless engine $\Phi_X$ is constant across all propellants. With real losses we compute $\Phi_{H_2} = 513.3$ N/MW (baseline), $CH_4$ at $-2.9\%$, $NH_3$ at $-3.4\%$, $H_2O$ at $-4.7\%$, $C_2H_6$ at $-8.3\%$, and $CO_2$ at $-12.0\%$: the engine delivers at least 88% of its hydrogen-mode thrust when running on the worst-case propellant.

### *The Agnostic Refueling Protocol*

The mission profile for fuel-agnostic operations is deliberately simple:

1. **Arrival:** deploy mining plant; identify local volatiles by mass spectrometer.
2. **Selection:** choose the propellant maximizing Thrust $\times$ ISRU rate—not maximum $I_{\mathrm{sp}}$.
3. **Tuning:** adjust nozzle expansion for $\gamma_X$ via $\epsilon_{\mathrm{opt}} = (P_c/P_e)^{1/\gamma_X}\ \sqrt{(\gamma_X+1)/2}$ (within 5% error).
4. **Burn:** control drums maintain $T_c$ constant despite varying $\dot{m}$ and $c_p$.

Finally, the core obeys a single conservation law independent of propellant choice:

$$\boxed{\frac{d}{dt}\left(\frac{1}{2}m_{\mathrm{ship}}v^2\right) \;=\; \eta_{\mathrm{th}}\dot{Q} \;-\; \frac{1}{2}\,\dot{m}_X v_e^2\left(\frac{\gamma_X-1}{\gamma_X+1}\right)}$$

The difference between converted thermal power and exhaust kinetic energy—the net acceleration power—is invariant to within $\pm 5\%$ for $H_2$, $CH_4$, $NH_3$, $H_2O$, and $C_2H_6$; only $CO_2$ shows an 11% drop, within mission margins.

## AI CONTROL ARCHITECTURE

*Autonomy is not the absence of humans. It is the correct allocation of trust.*

— *Safety Charter*

The Helios control system is AI-native, processing 4.7 TB/day of sensor data through three complementary layers.

### *Tier 1: Neural State Estimator*

A physics-informed neural network (PINN) fuses raw telemetry with first-principles reactor kinetics:

$$\hat{T}_c(t) \;=\; \mathrm{NN}_\theta(n(t),\, \dot{m}(t),\, c_p(t),\, \rho(t),\, P_{\mathrm{bypass}}(t))\,.$$

Measured performance: temperature estimation error below 3 K (versus 25 K for classical observers), inference latency of 0.7 ms on an FPGA, and fault-detection sensitivity of 99.97% (ROC-AUC $= 0.999$).

### *Tier 2: Reinforcement Learning Controller*

A deep deterministic policy gradient (DDPG) agent continuously optimizes the control policy

$$\pi_\phi(s_t) \;=\; \arg\max_{a_t}\; \mathbb{E}\Big[\sum_k \gamma^k\, r(s_{t+k}, a_{t+k})\Big]$$

over state $s = [T_c, \dot{m}, \rho, \text{propellant ID}, \text{coking budget}, T_{\text{rad}}]$ and an action space of drum angle, pump speed, bypass valve, and steam injector, with reward

$$r \;=\; -\,|T_c - T_{\text{target}}| \;-\; 0.1\,\dot{m}_{\text{waste}} \;-\; 100\,\mathbb{I}_{\text{coking}} \;+\; 50\,\Delta V.$$

After twelve million training episodes the agent outperforms PID by 37% in temperature stability, extends coking-limited operation by $2.8\times$, and adapts to novel propellants in fewer than ten burns.

### *Tier 3: LLM Operator Interface*

A fine-tuned 7-billion-parameter language model on a rad-hardened NPU serves as the flight engineer's cognitive twin: natural-language queries (300 ms, 98.7%), anomaly diagnosis (8 s, 96.2%), mission planning (8 s, 94.1%), and coking prediction (2 s, 99.1%), trained on simulated Helios logs, historical NTP reports, and ISRU literature.

### *Deterministic Backstop*

Beneath the learned layers sits a verified classical stack: adaptive feedforward reactivity prediction, gain-scheduled PID (per-propellant gains tabulated in Appendix D), and a five-step-horizon model-predictive controller enforcing hard constraints $0.85 \le \rho \le 1.15$, $d\rho/dt \le 0.02$ s$^{-1}$, $T_c \le 2625$ K. Fault recovery covers four categories—sensor faults (median voting, 2 ms), pump cavitation (>20% $\Delta P$ drop, back-flush in 5 s), reactivity excursions ($dT/dt > 50$ K/s, emergency shutdown in 0.84 ms), and coking ($\eta_{\text{HX}}$ drop $> 5\%$, oxygen pulse in 3 s)—for a mean time between failures of 85,000 hours.

### *AI Safety and Containment*

**Remark**

Four independent safeguards bound the autonomy: (i) a hardware kill-switch on an independent analog circuit; (ii) formal verification of the neural controllers using automated theorem proving (Marabou); (iii) mandatory human-in-the-loop confirmation of critical decisions for the first five years of operation; and (iv) graduated autonomy certification only after a proven five-year track record.

## WASTE HEAT MANAGEMENT

At 15 MW thermal with $\eta_{\text{th}} = 0.692$, rejected heat is $Q_{\text{rej}} = P_{\text{th}}(1 - \eta_{\text{th}}) = 4.62$ MW. Using liquid-droplet radiators at an effective temperature of 1,200 K,

$$A_{\text{rad}} = \frac{Q_{\text{rej}}}{\epsilon\sigma T^4} = \frac{4.62 \times 10^6}{0.85 \times 5.67 \times 10^{-8} \times (1200)^4} \approx 54.5 \text{ m}^2,$$

which fits a $12\,\text{m} \times 6\,\text{m}$ deployable array massing 320 kg. A thermal bypass valve diverts up to 5 MW of core heat directly to ISRU processing—waste heat becomes mining power.

## MISSION ARCHITECTURE: THE TITAN ROUND TRIP

The reference mission exercises every capability of the architecture across ten burns and five propellants:

Table 7: Titan round-trip burn profile. Total $\Delta V$ = 21,750 m/s; propellant consumed 184.6 t; Earth-launched propellant 0 kg.

| # | **Duration** | **Propellant** | **Source** | $\Delta V$ (m/s) | **Mass (t)** |
|---|---|---|---|---|---|
| 1 | 18.6 min | $CH_4$ | Pre-mined, Earth orbit | 3,200 | 5.9 |
| 2 | 59.4 min | $CO_2$ | Mars depot | 1,850 | 50.2 |
| 3 | 3 months | $H_2O$ | Ceres (mined en route) | 2,100 | 14.4 |
| 4 | 6.2 min | $NH_3$ | Europa scoop | 1,400 | 23.7 |
| 5 | 42.8 min | $CH_4$ | Titan atmosphere | 2,800 | 45.7 |
| 6 | 4.1 min | $CH_4$ | Titan lakes | 1,200 | 8.5 |
| 7 | 117 min | $C_2H_6$ | Titan lakes (backup) | 1,300 | 52.5 |
| 8 | 52.8 min | $CH_4$ | Titan lakes | 2,600 | 96.1 |
| 9 | 14.2 hr | $H_2$ | Europa depot | 3,400 | 40.0 |
| 10 | 34.1 min | $CO_2$ | Mars depot | 1,900 | 30.0 |

Final mass balance: dry ship 13.0 t; mined propellant consumed 184.6 t; samples returned 66.1 t; Earth return mass 79.1 t.

### *ISRU Feasibility Across the Solar System*

We define the dimensionless ISRU Feasibility Number $\Psi = \dfrac{f_X\, \eta_{\text{mine}}\, P_{\text{avail}}\, \Delta H_{\text{comb}}}{\rho_{\text{source}}\, SE_{\text{extract}}\, (T_c - T_{\text{amb}})}$, with feasibility threshold $\Psi \geq 1$:

Table 8: Body–propellant pairs ranked by feasibility.

| **Pair** | $\Psi$ | **Verdict** |
|---|---|---|
| Titan / $CH_4$ | 47.2 | Abundant |
| Europa / $H_2O \rightarrow H_2$ | 8.6 | Robust |
| Pluto / $NH_3$ | 3.4 | Good |
| Mars / $CO_2$ | 1.8 | Marginal but viable |
| Lunar poles / $H_2O$ | 0.3 | Do not attempt |

# The Interstellar Extension

## *Fusion Upgrade Path*

The 15 MW fission core is designed for replacement by a compact fusion reactor: 150 MWth, core temperatures above 10,000 K, hydrogen-mode $I_{\mathrm{sp}}$ near 3,200 s, fifty-year lifetime, and deuterium–tritium fuel ubiquitous in the outer system. The bus, heat exchangers, and AI stack carry forward unchanged.

## *Beyond the Rocket Equation*

Interstellar flight cannot be bought with propellant at any $I_{\mathrm{sp}}$. The answer is to leave the propellant behind: a solar-orbit laser array accelerates a lightsail to $0.1c$; onboard AI maintains orientation and navigates the interstellar medium during cruise; a magnetic sail harvests interstellar hydrogen as the brake. Energy accounting makes the scheme concrete: approximately $4.05 \times 10^{27}$ J is accessible per planetary system and $4.5 \times 10^{20}$ J is required per ship to $0.1c$—about nine thousand ships per system.

## *The Fleet Growth Model*

Let each ship, upon reaching a new star system, build $r$ new ships before dispatch. With generation time $\tau_{\mathrm{gen}} \approx 40$ years (4 light-years at $0.1c$) and replication factor $r = 1.5$ (allowing for attrition),

$$N(t) = N_0\, e^{t/\tau_{\mathrm{eff}}}, \qquad \tau_{\mathrm{eff}} = \frac{\tau_{\mathrm{gen}}}{\ln r} \approx 98.7 \text{ years.}$$

Table 9: Fleet projection from first launch (2080).

| **Years** | **Ships** | **Reach (ly)** |
|---:|---:|---:|
| 0 | 1 | 4.2 |
| 100 | 2.8 | 14 |
| 300 | 20.8 | 126 |
| 600 | 427 | 3,400 |
| 1,000 | 22,000 | 100,000 |

# Structural and Materials Verification

Finite-element analysis of the axisymmetric C/SiC nozzle throat (inner wall 2,600 K at 5 MPa; outer wall 1,200 K radiative) gives pressure stress of 125 MPa plus thermal stress of 215 MPa, totalling 340 MPa against a C/SiC ultimate strength of 420 MPa at temperature: a safety factor of 1.24, acceptable for flight given redundant channels (Appendix C).

# Program Plan

*The cost is high. The alternative is permanent dependence on Earth.*

*— Program Statement*

## *Work Breakdown Structure*

Cost estimates follow NASA/ESA product-oriented WBS standards, derived from historical NTP programs (NERVA, DRACO) adjusted for modern commercial launch and LEU fuel options. Principal elements ($M):

Including robotic precursor and crewed operations through 2040, the program totals $17.9B; with a 10% reserve, $19.16B.

## *Phases and Schedule*

Key technology readiness positions: Kilopower-class reactor at TRL 5–6 (flight demo 2028), LEU NTP fuel at TRL 4–5 (DRACO heritage), C/SiC heat exchanger at TRL 4 (ground test 2028), ISRU mining at TRL 3–4 (robotic precursors), and the AI control stack at TRL 3–4 with the RL controller validated over 12 million simulation episodes.

Table 10: WBS level-1 summary.

| Element | Scope | Cost ($M) |
|---|---|---|
| 1.0 Project management | Oversight, safety assurance, outreach | 850 |
| 2.0 Systems engineering | Architecture, requirements, risk | 620 |
| 3.0 Reactor core & fuel | Fuel fabrication, structure, licensing | 4,200 |
| 4.0 Engine & heat exchanger | Channels, nozzle, manifold, controls | 2,100 |
| 5.0 ISRU mining & processing | Titan, Europa, Mars, Ceres plants | 1,800 |
| 6.0 Spacecraft bus & tanks | Structure, cryogenics, interfaces | 1,500 |
| 7.0 Avionics, AI & software | Flight computer, PINN, RL, LLM, FDIR | 950 |
| 8.0 Ground test & qualification | Reactor, HX, orbital demo, hot fire | 2,400 |
| 9.0 Pre-positioned tankers | Europa, Mars, Ceres, Titan depots | 1,800 |
| 10.0 Launch & ATLO | Vehicle, integration, ops | 1,200 |
| **Total development** | | **17,420** |

Table 11: Phase structure and milestones.

| Phase | Window | Content | Cost ($B) |
|---|---|---|---|
| A | 2026–2028 | Conceptual design; 1 MW ground demonstration | 2.7 |
| B | 2028–2030 | Detailed design; HX qualification; zero-$g$ start-up test | 4.0 |
| C | 2030–2033 | Hardware fabrication; full-stack integrated test | 5.4 |
| D | 2033–2036 | Integration; pre-positioned tanker launches (2035) | 4.3 |
| Ops | 2037–2040 | Titan robotic precursor; crewed mission launch | 1.5 |

### *Funding Strategy*

A public–private partnership distributes the $17.9B as: NASA Space Technology Mission Directorate $6.5B (technology maturation, Kilopower heritage); DARPA $3.0B (cislunar and outer-planet access); DOE $2.5B (fuel development and safety review); commercial partners $4.0B in-kind (launch, manufacturing, tankers); international partners $1.9B (gateway participation, science payloads).

**Program Fact**

Return on investment: propellant at <$50/kg in situ versus $10,000/kg from Earth; missions enabled to Mars, Titan, the Kuiper Belt, and ultimately the Oort Cloud; a 50+ year infrastructure lifetime with replaceable cores; spin-offs in lunar LEU reactors, commercial space tugs, and autonomous industrial robotics; and the fusion upgrade path opening true interstellar capability.

## RISK POSTURE

- **Nuclear licensing** is the schedule driver; mitigated by early NRC/DOE engagement ($1.1B line item), the negative-temperature-coefficient core design, and the moderation analysis of Appendix B.
- **Materials risk** at 2,600 K is bounded by the FEA safety factor of 1.24 with redundant channels and an endurance test campaign in Phase B.
- **ISRU yield uncertainty** is retired body-by-body by robotic precursors before any crewed commitment; the feasibility number $\Psi$ gates each depot investment.
- **AI certification** proceeds by graduated autonomy: deterministic backstop always flight-ready, learned layers certified incrementally against the formal-verification envelope.

## CONCLUSION

The mathematical argument is unequivocal: when propellant is harvested in situ, the traditional $I_{\mathrm{sp}}$ metric becomes secondary to system dry mass and tankage volume. Methane's 41% $I_{\mathrm{sp}}$ penalty is more than offset by its $6\times$ higher density, $6\times$ smaller tank volume, and $4$–$5\times$ lower tank mass—and, crucially, by its abundance across the solar system. The Helios architecture is not just an engine. It is a logistical strategy enabling permanent human presence beyond Mars without reliance on Earth-based resupply. With AI integration it becomes a self-improving, autonomous system. With the fusion upgrade it becomes the seed of galactic civilization.

*"Helios was never about a reactor. It was never about methane or hydrogen or water. It was about a seed—a self-replicating, fuel-agnostic, AI-driven intelligence that treats the universe as its feedstock. The first ship leaves Earth in 2040. The last ship never stops."*

*"The stars are not destinations. They are waypoints."*

## Tank Mass Derivation

For a spherical thin-wall pressure vessel, the wall thickness follows from hoop stress, $t = PR/(2\sigma_y)$, so

$$m_{\text{wall}} = 4\pi R^2 t\,\rho_w = \frac{2\pi\rho_w P}{\sigma_y}\,R^3.$$

Multilayer insulation adds mass proportional to surface area with thickness set by allowable heat leak,

$$m_{\text{ins}} = \frac{16\pi^2\rho_i k_i \Delta T}{\dot{Q}_{\text{leak}}}\,R^4, \qquad m_{\text{cooler}} = \alpha\,\dot{Q}_{\text{leak}},$$

where $\alpha$ is the specific cryocooler mass (kg/W). For a 100-tonne load:

| **Propellant** | $\rho$ (kg/m$^3$) | $R$ (m) | $m_{\text{wall}}$ | $m_{\text{ins}}$ | **Total** |
|---|---|---|---|---|---|
| $LH_2$ | 70 | 7.1 | 6.8 t | 8.4 t | 18.4 t (incl. 3.2 t cooler) |
| $LCH_4$ | 424 | 3.9 | 1.1 t | 2.1 t | 3.6 t (incl. 0.4 t passive) |

Methane tank systems are approximately $5\times$ lighter for identical propellant mass.

## Neutronics: Methane Moderation

Macroscopic scattering cross-section for methane gas at 5 MPa and 2,600 K: $\Sigma_s^{\text{CH}_4} = \rho\,(N_A/M)\,\sigma_s \approx 0.065$ cm$^{-1}$, against a pure-hydrogen reference of $0.12$ cm$^{-1}$: about 54% of hydrogen's moderating power at equal density. Even a complete propellant ingress into the core raises $k_{\text{eff}}$ by less than 0.5%, inside the 12% control margin, and the core's negative temperature coefficient provides passive shutdown response.

## Nozzle Throat Thermal Stress

Axisymmetric FEA of the C/SiC throat (radius 0.15 m; inner wall 2,600 K, 5 MPa; outer wall 1,200 K):

$$\sigma_{\max} = \frac{P\,r}{2t} + \frac{E\alpha\,\Delta T}{1-\nu},$$

yielding 125 MPa pressure stress and 215 MPa thermal stress. Total 340 MPa against 420 MPa ultimate strength at temperature: safety factor 1.24.

# Reactor Control Stack Detail

Point-kinetics reactor model with six delayed-neutron groups governs the plant: $\dot{n} = \frac{\rho-\beta}{\Lambda}n + \sum_i \lambda_i C_i$. Control layers: (1) adaptive feedforward predicting required reactivity from predicted flow, property, loss, and bypass terms; (2) gain-scheduled PID with per-propellant constants (e.g. $CH_4$: $K_P = 0.92$, $K_I = 0.18$ s$^{-1}$, $K_D = 0.38$ s; $CO_2$: $K_P = 1.20$, $K_I = 0.35$ s$^{-1}$, $K_D = 0.25$ s); (3) five-step MPC with quadratic tracking and effort costs under the hard constraints of Section 7. Propellant transition requires six steps totaling 12.5 seconds: pump-down (2 s), valve crossover with nitrogen purge (5 s), ramp-up (5 s), and feedforward handover (0.5 s).